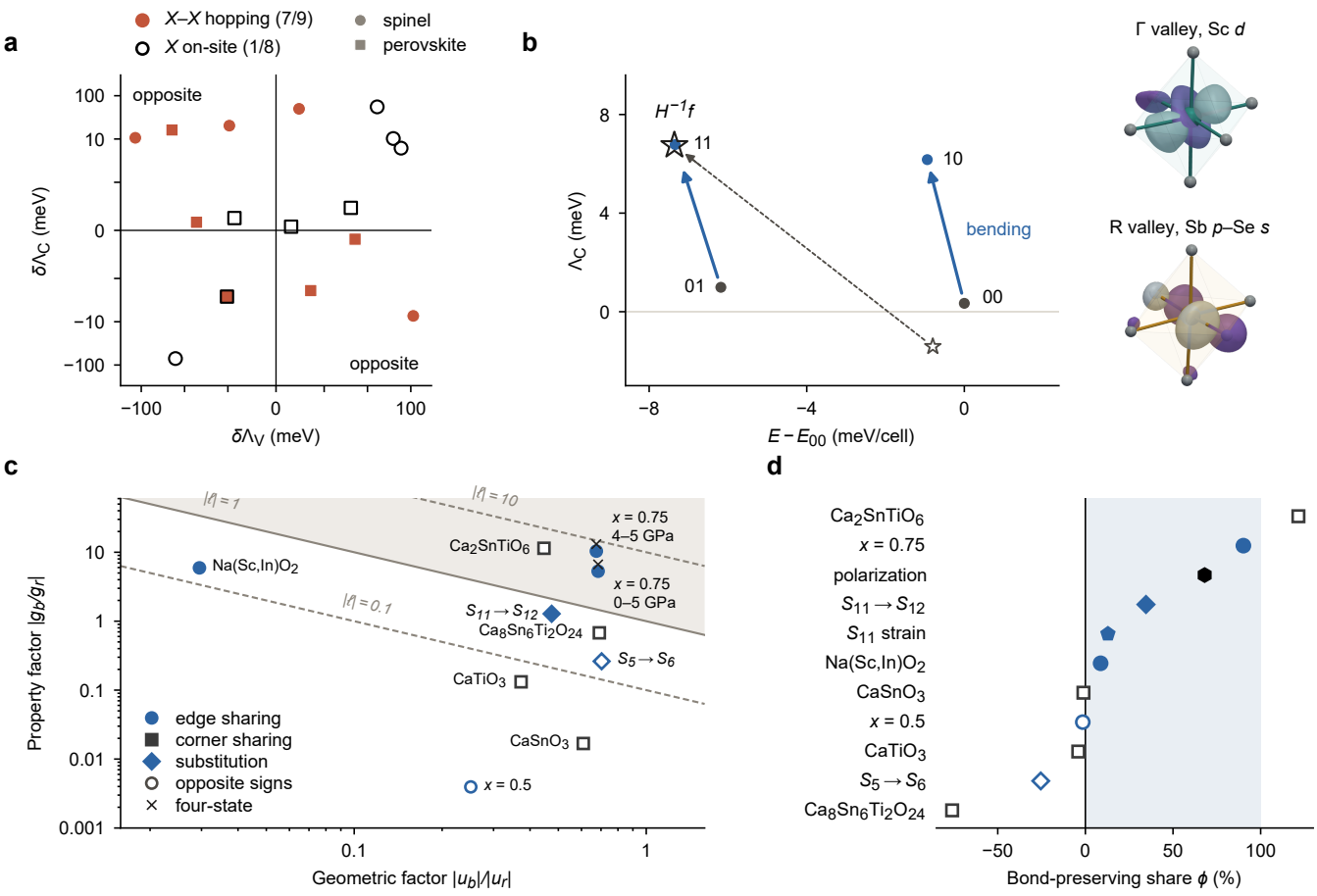

a
X–X hopping (7/9)
X on-site (1/8)
spinel
perovskite
opposite
opposite
δΛC (meV)
δΛV (meV)
b
H⁻¹f
11
10
bending
01
00
ΛC (meV)
E − E00 (meV/cell)
Γ valley, Sc d
R valley, Sb p–Se s
c
|f| = 1
|f| = 10
|f| = 0.1
Ca2SnTiO6
x = 0.75
4–5 GPa
x = 0.75
0–5 GPa
Na(Sc,In)O2
S11 → S12
Ca8Sn6Ti2O24
S5 → S6
CaTiO3
CaSnO3
x = 0.5
edge sharing
corner sharing
substitution
opposite signs
four-state
Property factor |gb/gr|
Geometric factor |ub|/|ur|
d
Ca2SnTiO6
x = 0.75
polarization
S11 → S12
S11 strain
Na(Sc,In)O2
CaSnO3
x = 0.5
CaTiO3
S5 → S6
Ca8Sn6Ti2O24
Bond-preserving share ϕ (%)